

1. Discourse, Reference Resolution & Text Coherence

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PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/17.py"
REFERENCE RESOLUTION
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Original Text:
Ravi met Arun at the library. He borrowed a book and later returned it.

Resolved Text:
Ravi met Arun at the library. Ravi borrowed a book and later returned book.

Coreference Chains:
Ravi --> He
book --> it

TEXT COHERENCE ANALYSIS
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Sentence 1: The roads were flooded after heavy rainfall.
Sentence 2: Therefore, schools were closed for the day.
Relation: Cause-Effect

Sentence 1: Therefore, schools were closed for the day.
Sentence 2: Students attended classes online.
Relation: Sequence
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```

2. Dialog and Conversational Agents, Discourse Planning & Surface Realization

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PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/18.py"
DIALOGUE ACT RECOGNITION
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Speaker: User
Utterance: Can you book a train ticket for me?
Dialogue Act: Request

Speaker: Agent
Utterance: Sure, where would you like to travel?
Dialogue Act: Question

Speaker: User
Utterance: I want to go to Chennai.
Dialogue Act: Inform

Speaker: Agent
Utterance: Your ticket has been booked.
Dialogue Act: Confirmation / Action

DISCOURSE PLAN
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1. Identify user goal
2. Collect missing information
3. Store user-provided information
4. Complete task and confirm result
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3. Machine Translation, Discourse Structure, Predicate Logic & Interpretation

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2. PREDICATE LOGIC REPRESENTATION
PLAY(boy, football) AND TENSE(PRESENT) AND ASPECT(PROGRESSIVE)

3. INTERLINGUA REPRESENTATION
event: PLAY
agent: BOY
theme: FOOTBALL
tense: PRESENT
aspect: PROGRESSIVE

4. TRANSLATION CANDIDATES
बच्चा फुटबॉल खेल रहा है।
फुटबॉल बच्चा खेल रहा है।
बच्चा खेल रहा है फुटबॉल।

5. STATISTICAL SCORES
बच्चा फुटबॉल खेल रहा है। --> Combined Score: 0.955
फुटबॉल बच्चा खेल रहा है। --> Combined Score: 0.550
बच्चा खेल रहा है फुटबॉल। --> Combined Score: 0.400

FINAL TRANSLATION
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बच्चा फुटबॉल खेल रहा है।
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```